Saturday — Student Practical Lab

Work through these exercises in the same order as the lesson. Predict outputs and explain what your code is doing. Do not consult the answer key before attempting each problem.

Student: _____ Date: _____ Class: _____

Lab submission instructions

Write the program in Python, test at least two inputs, and record the result. For each task, describe one concept used (such as variables, input, list, loop, function).

Lab 01 — Quick Check — Predict the Output

Without running Python, what will be printed?

```
print("Python", 3)
print(6 + 4)
print("6 + 4")
```

Hint: Is the expression written inside quotes?

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 02 — Practice 1A — Student Profile

Create variables for your name, age, GPA, and whether you are enrolled. Print all four values.

Hint: Use str, int, float and bool values.

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 03 — Practice 1B — Character Profile

Make a character profile with first name, last name, age, height and a Boolean value. Ask the user for a nickname and print the profile.

Hint: Inspired by the Tony Stark exercise in the supplied notes.

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 04 — Practice 2A — Total & Average Bill

Ask for the prices of three products. Show the total cost and average price.

Hint: Use float(input(...)); average = total / 3.

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 05 — Practice 2B — Name Checker

Ask for a superhero name. Print whether the name begins with S or s.

Hint: Use name.lower().startswith("s").

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 06 — Practice 3A — Predict True/False

Predict each result, then test your answers:

```
print(10 > 5 and 3 > 7)
print(10 > 5 or 3 > 7)
print(not (3 == 3))
```

Hint: Evaluate each comparison first.

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 07 — Practice 3B — Simple Grade Checker

Ask the user to enter marks. Print "Pass" if marks are 50 or higher; otherwise print "Fail".

Hint: Convert input using int() and write if/else.

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 08 — Calculator — Student Challenge

Extend your calculator to support remainder (%) and power (**). Also test what happens when the user enters 0 as the divisor.

Hint: Challenge follows the supplied PDF calculator exercise.

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 09 — Practice 4 — Loops

A. Print odd numbers from 1 to 20.
B. Print the multiplication table of 7.
C. Print 1 to 10 but skip 5.

Hint: Try range(start, stop, step) first; use continue for C.

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 10 — Practice 5A — List of Names

Create a list of three student names. Add another name, change the first name, and print each name using a loop.

Hint: append(), list[0] = ..., for name in names

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 11 — Practice 5B — Collections

A. Remove duplicate student roll numbers.
B. Store the marks of three subjects in a dictionary.
C. Print the marks for one subject.

Hint: Use set(), a dictionary literal, and dictionary["key"].

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 12 — Practice 5C — Find an Employee

Create a small dictionary where each employee ID maps to a name. Ask for an ID and display the name, or say "Not found".

Hint: Use a dictionary and the in operator.

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 13 — Practice 6A — Write Your Functions

A. Write check_odd_even(number).
B. Write average_marks(scores) to return the mean.
C. Call both functions with sample values.

Hint: Odd/even uses %. Average uses sum() / len().

Test inputs: _____

Observed output: _____

Concept learned: _____

Lab 14 — Guessing Game — Student Improvements

1. Count how many guesses the player makes.
2. Ask if the player wants to try again.
3. Change the range from 1–10 to 1–20.

Hint: Starter task: add attempts = 0 before the loop and attempts += 1 inside.

Test inputs: _____

Observed output: _____

Concept learned: _____

Reflection

1. Which topic was easiest and why? 2. What was the most common error you encountered? 3. Which function can you now write without looking at notes?